POLITECNICO DI TORINO · Master Degree in Data Science

Official Examination Session 1

Date: June 30, 2025 | **Time Allotted:** 1 Hour 25 Minutes | **Total Points:** 16.00 Points

Session Scope
Summer Examination Session

§1 Exam Session 1 — June 30, 2025

Session Overview: This exam comprises 6 questions covering fundamental taxonomy, mechanistic interpretability, concept-based versus saliency-based explainability, adversarial attacks, cooperative game theory foundations of SHAP, and an in-depth design problem for medical image diagnosis (total: 16.00 points).

Question 1: Model-Agnostic Explainability — MCQ — 1.00 Point

What does "model-agnostic" mean in the context of explainability?

a. The explanation method can be applied to any machine learning model.

b. The explanation method is specific to a particular model type.

c. The explanation method ignores the input data.

d. The explanation method is only applicable to neural networks.

✓ Executive Solution Summary

The correct answer is **a**. A *model-agnostic* explanation method treats the machine learning model as a black box $f : \mathcal{X} \to \mathcal{Y}$, requiring only the ability to query the model with arbitrary inputs and observe the corresponding outputs. It makes zero assumptions about the internal architecture, weights, or training procedure of the underlying model.

➤ Systematic Analysis of Multiple Choice Options

(a) [CORRECT] The explanation method can be applied to any machine learning model.

Model-agnostic methods operate externally by perturbing inputs $x \in \mathcal{X}$, observing predictions $f(x)$, and constructing surrogates or calculating empirical differences. They function uniformly whether f is a deep neural network, random forest, gradient-boosted tree, support vector machine, or an ensemble.

(b) [INCORRECT] The explanation method is specific to a particular model type.

This defines **model-specific** methods (e.g., TreeSHAP for tree ensembles, Grad-CAM or Integrated Gradients for differentiable neural networks), which directly inspect internal parameters, activations, or gradients.

(c) [INCORRECT] The explanation method ignores the input data.

Every explanation method fundamentally depends on input data instances to either

evaluate predictions locally (x) or assess marginal variations across a dataset ($\mathcal{D}$). Ignoring input data is nonsensical.

(d) [INCORRECT] The explanation method is only applicable to neural networks.
This describes deep-learning-specific techniques (such as backpropagation-based saliency maps, layer-wise relevance propagation, or neuron activation analysis), which represent the exact opposite of model-agnosticism.

✍ Comprehensive Technical Walkthrough

In the taxonomy of Explainable AI (XAI), the criterion of **generalizability** divides explanation techniques into two fundamental classes:

Dimension	Model-Agnostic	Model-Specific
Access Level	Black-box oracle access ($x \mapsto f(x)$)	White-box access (θ, $\nabla_x f$, activations)
Portability	Universal across all model families	Restricted to specific mathematical structures
Key Advantage	Fair benchmarking across heterogeneous models	High computational efficiency, exact solutions
Canonical Examples	LIME, KernelSHAP, PFI, PDP, LORE	TreeSHAP, Grad-CAM, Integrated Gradients

Formally, let $\mathcal{M}$ be the set of all predictive models. A model-agnostic explainer $\mathcal{E}$ is an operator:

$$\mathcal{E} : (f, x, \mathcal{D}_{bg}) \mapsto \mathbf{e} \quad \forall f \in \mathcal{M} \tag{1}$$

where f is treated strictly as an evaluation oracle, x is the query instance, $\mathcal{D}_{bg}$ is an optional background dataset, and $\mathbf{e}$ is the resulting explanation (e.g., feature attribution vector or decision rule).

★ Examiner Feedback & Common Pitfalls (Prof. Eliana Pastor)

A recurring misconception in previous exam sessions is conflating *model-agnostic* post-hoc methods with *inherently interpretable* models (explainable-by-design). Model-agnostic refers strictly to the **applicability scope of the explainer**, not to whether the underlying predictive model itself is interpretable.

➥ Course Theory Mapping (Notes & Slides)

NOTES/XAI_Theory.pdf – Chapter 2: "Explainable AI: Taxonomy", Section: "Generalizability of Explainability" & "Advantages of Model-Agnostic Solutions" (pp. 15–18).

¶ Alternative Perspectives & Methodological Variants

While model-agnostic methods guarantee maximal flexibility, their model-specific counterparts leverage internal structures to achieve dramatic speedups: for example, TreeSHAP computes exact Shapley values in low-order polynomial time $\mathcal{O}(TLD^2)$, whereas KernelSHAP requires exponential $\mathcal{O}(2^M)$ black-box evaluations or Monte Carlo approximations.

Question 2: Mechanistic Interpretability Core Assumptions MCQ 1.00 Point

Which of the following statements is **FALSE** regarding mechanistic interpretability:

a. Techniques in mechanistic interpretability often involve identifying specific neurons or patterns responsible for particular behaviors.

b. All layers in a deep neural network encode the same types of interpretable features through mechanistic analysis.

c. Mechanistic interpretability aims to reverse-engineer neural networks into human-understandable components like circuits and algorithms.

d. Mechanistic interpretability has shown promising results in understanding how models internally represent features like syntax or modular arithmetic.

✓ **Executive Solution Summary**

The correct answer is **b** (the FALSE statement). Deep neural networks exhibit a pronounced **hierarchical representation**: early layers encode low-level primitive features (edges, textures, Gabor filters), intermediate layers assemble these into motif combinations, and deep layers encode high-level abstract semantic representations. Thus, different layers encode fundamentally distinct types of features.

➤ **Systematic Analysis of Multiple Choice Options**

(b) [CORRECT] All layers in a deep neural network encode the same types of interpretable features through mechanistic analysis.

This is provably **FALSE**. Hierarchical feature learning is a foundational tenet of deep architectures. In CNNs, early layers compute Gabor-like directional filters, middle layers compute textures and parts, and late layers compute object detectors. In Transformers, early layers attend to local n-grams and syntax, while deeper layers implement semantic induction heads and long-range algorithmic reasoning.

(a) [INCORRECT] Techniques in mechanistic interpretability often involve identifying specific neurons or patterns responsible for particular behaviors.

This statement is **TRUE**. Mechanistic interpretability frequently isolates individual units, directions in latent space, or specific attention heads (e.g., indirect object identification heads) that causally drive specific model behaviors.

(c) [INCORRECT] Mechanistic interpretability aims to reverse-engineer neural networks into human-understandable components like circuits and algorithms.

This statement is **TRUE**. It defines the primary research mission of mechanistic interpretability: deconstructing black-box parameters into discrete computational subroutines (circuits).

(d) [INCORRECT] Mechanistic interpretability has shown promising results in understanding how models internally represent features like syntax or modular arithmetic.

This statement is **TRUE**. Groundbreaking studies have deciphered how transformers implement modular addition through discrete Fourier transforms (grokking) and track grammatical syntax trees.

✍ **Comprehensive Technical Walkthrough**

Mechanistic interpretability treats neural networks not as statistical black boxes, but as compiled computer programs. The discipline is structured around the **Three Speculative Claims** formulated by Olah et al.:

Claim 1. Features: The atomic units of neural networks are *features* (directions in activation space) corresponding to meaningful properties of the input.

Claim 2. Circuits: Features are connected by weights to form *circuits*, which compute sophisticated subroutines (algorithms).

Claim 3. Universality: Across different models, random initializations, and training runs, similar features and circuits independently emerge when optimizing for the same tasks.

The progression of representations across layers $l \in \{1, \ldots, L\}$ obeys strict compositional hierarchy:

$$h^{(l)}(x) = \sigma\left(W^{(l)} h^{(l-1)}(x) + b^{(l)}\right) \tag{2}$$

Because each layer applies a non-linear transformation over the linear combinations of previous-layer features, the semantic complexity of features strictly grows with depth:

$$\text{Semantic Complexity}\left(\mathcal{F}^{(1)}\right) \prec \text{Semantic Complexity}\left(\mathcal{F}^{(l)}\right) \prec \text{Semantic Complexity}\left(\mathcal{F}^{(L)}\right) \tag{3}$$

Claiming that all layers encode the "same types" of features violates the core mathematical foundation of deep representation learning.

★ **Examiner Feedback & Common Pitfalls (Prof. Eliana Pastor)**

Students frequently overlook the negative phrasing of the question (identifying the **FALSE** statement). Furthermore, some confuse the property of *universality* (features recurring across different models) with the false claim that features are identical across different layers of the *same* model.

➥ **Course Theory Mapping (Notes & Slides)**

NOTES/XAI_Theory.pdf – Chapter 11: "Mechanistic Interpretability", Sections: "The Three Speculative Claims", "Feature Hierarchy", and "Mechanistic Interpretability in LLMs" (pp. 155–162).

¶ **Alternative Perspectives & Methodological Variants**

Representational Similarity Analysis (RSA) and Centered Kernel Alignment (CKA) quantitatively confirm that feature representations across disparate network depths have low mutual alignment, verifying mathematically that distinct layers perform qualitatively different stages of feature computation.

Question 3: Limitations of Saliency Maps vs. Concept-Based Explainability Open Inquiry 3.00 Points

What are the main limitations of standard explanation methods (like saliency maps)? How do concept-based explainability methods try to solve them?

✓ **Executive Solution Summary**

Standard explanation methods (such as pixel-level saliency maps) suffer from five critical limitations: (1) the **cognitive/semantic gap** (pointing *where* a model looks rather than *how* or *why* it reasons), (2) **class insensitivity** and visual artifacts (passing off edge detection as reasoning), (3) **sign/directional ambiguity**, (4) **gradient noise and saturation**, and (5) **passivity** (inability to support expert intervention).

Concept-based explainability resolves these deficiencies by shifting the unit of explanation from low-level pixels to high-level, human-understandable **concepts** ($c \in \mathcal{C}$), implemented either **post-hoc** (e.g., TCAV via directional derivatives in activation space) or **explainable-by-design** (e.g., Concept Bottleneck Models $x \to c \to y$), enabling symbolic reasoning, auditable decision boundaries, and test-time concept intervention.

✍ **Comprehensive Technical Walkthrough**

1. Critical Limitations of Standard Saliency Methods

Standard gradient-based attribution methods calculate the sensitivity of a target class score $S_c(x)$ with respect to individual input pixels:

$$A_{\text{Vanilla}}(x) = \nabla_x S_c(x) = \frac{\partial S_c(x)}{\partial x} \tag{4}$$

While intuitive to visualize as heatmaps, they suffer from deep theoretical and operational failures:

L1. The Semantic / Cognitive Gap ("Where" vs. "Why"): A saliency map highlights spatial regions correlated with the prediction, but cannot convey the semantic logic or rule applied. For instance, highlighting a dog's ears does not clarify whether the classification was driven by shape, fur texture, breed markers, or an anomalous background artifact.

L2. Class Insensitivity & Sanity Check Failures (Adebayo et al.): Many popular saliency methods (e.g., Guided Backpropagation, Guided Grad-CAM) produce visually pleasing edge maps that are largely invariant to model parameters or class labels. When model weights are completely randomized, the saliency maps remain virtually identical, behaving as edge detectors rather than faithful explanations of decision logic.

L3. Identical Attributions for Correct and Erroneous Decisions: When a model misclassifies an input, saliency maps frequently highlight the exact same visual regions as they would for a correct prediction, failing to reveal the underlying causal mistake.

L4. Gradient Saturation and Local Instability: Due to non-linear activations (e.g., ReLU flat regimes), gradients can vanish ($\nabla_x S_c = 0$) even when a feature is causally decisive (saturation). Conversely, extreme local non-linearities cause high-frequency visual noise.

L5. Absence of Actionability and Intervention: Saliency maps are strictly passive observation tools. If a radiologist discovers that a model is attending to an imaging artifact, pixel heatmaps offer no direct mechanism to correct the model's internal reasoning.

2. Concept-Based Explainability Framework

Concept-based methods bridge the semantic gap by defining explanations in terms of user-friendly, high-level abstractions called **concepts** $\mathcal{C} = \{c_1, c_2, \ldots, c_k\}$ (e.g., *"striped"*, *"bone fracture"*, *"pedestrian crosswalk"*).

Paradigm	Post-Hoc Concept Methods (e.g., TCAV)	By-Design Concept Models (CBMs / CEMs)
Model Architecture	Unmodified black-box network $f(x)$	Constrained bottleneck: $x \xrightarrow{g} c \xrightarrow{f} y$
Explanation Mechanism	Directional derivatives along Concept Activation Vectors ($\mathbf{v}_C^l$)	Direct linear/interpretable weights from concepts to classes
Model Retraining	Not required; evaluates pre-trained latent space	Required; trained end-to-end with concept objectives
Intervention Capability	Passive verification and bias auditing	Active test-time concept intervention ($c_j \leftarrow c_j^*$)

A. Post-Hoc: Testing with Concept Activation Vectors (TCAV)

TCAV translates internal network activations at an intermediate layer l into human concepts without modifying the model:

1. **Define CAV ($\mathbf{v}_C^l$):** Collect positive examples of concept C and random negative examples. Train a linear classifier in activation space $\mathbb{R}^{d_l}$ to separate them; the normal vector pointing toward concept activations is the CAV $\mathbf{v}_C^l$.
2. **Compute Directional Derivative:** Quantify the sensitivity of class logit $h_k(x)$ to concept C:
$$S_{C,k,l}(x) = \nabla_{a_l} h_k(x) \cdot \mathbf{v}_C^l \tag{5}$$
3. **Calculate TCAV Score:** Fraction of class instances with positive sensitivity:
$$\text{TCAV}_{Q,k,l} = \frac{|\{x \in X_k : S_{C,k,l}(x) > 0\}|}{|X_k|} \tag{6}$$

B. Explainable-by-Design: Concept Bottleneck Models (CBMs)

A Concept Bottleneck Model explicitly forces the network's internal representation through a concept layer:
$$x \xrightarrow{g} \hat{c} \xrightarrow{f} \hat{y} \quad \text{where} \quad g : \mathcal{X} \to [0,1]^K, \quad f : [0,1]^K \to \mathcal{Y} \tag{7}$$

- **Interpretability:** If f is a generalized linear model, the prediction for class y is directly given by $\hat{y} = \sigma\left(\sum_{j=1}^{K} w_j \hat{c}_j + b\right)$, providing exact, faithful concept attributions $w_j \hat{c}_j$.
- **Human-in-the-Loop Intervention:** If the concept encoder $g(x)$ errs on concept j, a domain expert can overwrite the value $\hat{c}_j \leftarrow c_j^*$, directly updating $\hat{y}$ without retraining.
- **Training Strategies:**
  - *Independent:* Train $g(x) \to c^*$ and $f(c^*) \to y$ separately. Eliminates concept leakage, but task accuracy may suffer.
  - *Sequential:* Train $g(x) \to c^*$, freeze g, then train $f(g(x)) \to y$.
  - *Joint:* Train both simultaneously minimizing $\mathcal{L}_{\text{task}}(f(g(x)), y) + \lambda \mathcal{L}_{\text{concept}}(g(x), c^*)$. Balances accuracy and concept alignment, though risking concept leakage.

★ Examiner Feedback & Common Pitfalls (Prof. Eliana Pastor)

In past exams, students who received top marks were those who covered **both** post-hoc methods (TCAV) and explainable-by-design architectures (CBMs). A frequent deduction (-0.25 points) occurred when candidates focused exclusively on CBMs and ignored post-hoc concept extraction, or only cited a single limitation of saliency maps (-0.25 points).

➥ Course Theory Mapping (Notes & Slides)

NOTES/XAI_Theory.pdf – Chapter 5: "Gradient-based Explainability Methods" (pp. 65–78) & Chapter 7: "Concept-based Explainable AI" (pp. 92–110).

¶ Alternative Perspectives & Methodological Variants

Concept Embedding Models (CEMs) replace discrete scalar concepts with continuous concept embeddings $\mathbf{e}_c \in \mathbb{R}^d$, overcoming the accuracy drop of strict bottlenecks while retaining the ability to intervene on individual semantic concepts.

Question 4: Adversarial Attacks: Definition and Significance in Trustworthy AI — Open Inquiry — 2.50 Points

What is an adversarial attack? Why are adversarial attacks important?

✓ Executive Solution Summary

An **adversarial attack** is an intentional, malicious manipulation of an input instance via a human-imperceptible perturbation δ ($\|\delta\|_p \leq \epsilon$) designed to induce an arbitrary misclassification or steer the output toward an attacker-chosen target class:

$$f(x+\delta) \neq f(x) \quad \text{or} \quad f(x+\delta) = y_{\text{target}} \quad \text{s.t.} \quad \|\delta\|_p \leq \epsilon \tag{8}$$

Adversarial attacks are paramount in Trustworthy AI because they: (1) prove deep networks rely on brittle, non-robust statistical artifacts rather than true semantic understanding, (2) pose lethal physical threats in safety-critical deployments (autonomous driving, healthcare, airport screening), (3) exhibit cross-model transferability, (4) extend beyond vision to audio, NLP/LLM jailbreaking, and tabular domains, and (5) can be weaponized against explainability systems to generate deceptive explanations.

✍ Comprehensive Technical Walkthrough

1. Mathematical Formulation and Attack Taxonomy

Adversarial examples exploit the geometry of high-dimensional decision boundaries. For a classifier f_θ and an input (x, y), an untargeted adversarial perturbation δ is obtained by solving the constrained optimization:

$$\max_{\delta} \mathcal{L}(f_\theta(x+\delta), y) \quad \text{subject to} \quad \|\delta\|_p \leq \epsilon \tag{9}$$

where $\mathcal{L}$ is the loss function, and $\|\cdot\|_p$ is an L_p norm (L_∞ bounding maximum pixel change, L_2 bounding Euclidean energy, or L_0 bounding the number of altered pixels).

Dimension	Categories	Operational Characterization
Attacker Knowledge	White-Box vs. Black-Box	Full gradient access ($\nabla_x \mathcal{L}$) vs. Zero internal access (query-based or transfer-based)
Adversarial Goal	Targeted vs. Untargeted	Force a specific label y_{target} vs. Induce any erroneous label $y' \neq y$
Perturbation Scope	Instance-Specific vs. Universal	Optimized for single x vs. Single vector $\mathbf{v}$ fooling $\geq \delta\%$ of all dataset samples
Core Methods	FGSM, PGD, C&W	One-step gradient sign vs. Multi-step projected gradient vs. Optimization of margin loss

2. Why Adversarial Attacks Are Crucial to Trustworthy AI

Adversarial examples are not mathematical curiosities; they expose fundamental flaws in deep learning:

Reason 1. Disconnect Between Statistical Accuracy and True Robustness: Ilyas et al. proved that deep networks achieve high test accuracy by leveraging *non-robust features*—patterns highly predictive within i.i.d. datasets but imperceptible to humans and trivially brittle under adversarial shift. Thus, standard validation metrics offer a completely false sense of safety.

Reason 2. Real-World Physical Threat in Safety-Critical Systems: Adversarial examples successfully transition into physical reality via Expectation Over Transformation (EOT). Physical stickers on road signs fool autonomous vehicle perception (e.g., misclassifying a STOP sign as a 80 km/h speed limit sign), and printed adversarial camouflage clothing blinds surveillance object detectors.

Reason 3. Cross-Model Transferability: Perturbations crafted on a local surrogate white-box model often successfully fool completely different, unseen black-box production models, rendering proprietary security-by-obscurity defenses obsolete.

Reason 4. Vulnerability Across All Modalities: The threat extends to speech recognition (inaudible acoustic commands activating smart assistants), NLP and Large Language Models (universal adversarial suffixes, prompt injection, and jailbreaks that bypass alignment guardrails), and tabular financial systems.

Reason 5. Subversion of Explainability (Adversarial Explanations): Slack et al. demonstrated that post-hoc explainers like LIME and SHAP can be hijacked: an adversary can design a scaffolding wrapper that produces an equitable, benign explanation while the underlying model executes racist or fraudulent decision rules.

Reason 6. Regulatory Compliance and Legal Liability: Under the EU AI Act (Article 15), high-risk AI deployments are legally mandated to achieve technical robustness and demonstrable resilience against adversarial manipulations.

★ Examiner Feedback & Common Pitfalls (Prof. Eliana Pastor)

Prof. Eliana Pastor specifically penalizes answers that provide only a generic definition without detailing the multi-faceted reasons why attacks matter (-0.25 points deduction in past exams). A complete answer must explicitly discuss physical-world realization, cross-model transferability, non-vision modalities (LLMs/audio), and the security arms race.

➥ **Course Theory Mapping (Notes & Slides)**

NOTES/XAI_Theory.pdf – Chapter 12: "Adversarial Attacks and Defenses", Sections: "Why are Adversarial Attacks Important?", "Is the Threat Real?", and "Historical Evolution of Attacks" (pp. 175–189).

¶ **Alternative Perspectives & Methodological Variants**

Adversarial Training formulates model defense as a robust min-max game:

$$\min_{\theta} \mathbb{E}_{(x,y)\sim\mathcal{D}} \left[\max_{\|\delta\|_p \leq \epsilon} \mathcal{L}(f_\theta(x+\delta), y) \right] \tag{10}$$

providing empirical resilience, while **Randomized Smoothing** provides certified mathematical guarantees that predictions remain constant within an L_2 radius R.

Question 5: Core Idea of SHAP, Cooperative Game Theory, and Axiomatic Guarantees — Open Inquiry — 2.50 Points

Briefly describe the core idea behind SHAP (SHapley Additive exPlanations), and how it relates to Shapley values from cooperative game theory. Mention which kind of explanation representation it provides to the user. Outline one advantage of using SHAP for model interpretability.

✓ **Executive Solution Summary**

SHAP is a post-hoc, local, model-agnostic explanation framework that formulates feature attribution as a cooperative game. Individual features act as *players* in a coalition, and the model prediction serves as the *total payout*. SHAP assigns each feature its **Shapley value**—the weighted average of its marginal contributions across all possible feature subsets.
SHAP provides a **local feature attribution** representation (signed scalar contributions indicating how much each feature pushes the prediction above or below the expected base value). Its primary advantage is its **solid axiomatic foundation**: it is the *unique* attribution method satisfying Efficiency, Symmetry, Dummy Player, and Additivity/Monotonicity, while explicitly accounting for multi-feature interactions.

✍ **Comprehensive Technical Walkthrough**

1. Core Concept and Cooperative Game Theory Mapping

SHAP unifies additive feature attribution methods under the classic cooperative game theory framework established by Lloyd Shapley (1953). In this formulation:

- **Players ($N = \{1, 2, \ldots, M\}$):** The set of input features.
- **Coalition ($S \subseteq N$):** A subset of active features.
- **Characteristic Value Function $v(S)$:** The expected prediction of the model when only the features in S are known, marginalizing over absent features $\bar{S} = N \setminus S$:

$$v(S) = \mathbb{E}_{X_{\bar{S}}} \left[f(x_S, X_{\bar{S}}) \right] \tag{11}$$

- **Total Game Payout:** The deviation of the instance prediction $f(x)$ from the expected dataset base value $\mathbb{E}[f(X)]$.

The Shapley value ϕ_i of feature i represents its fair, marginal contribution averaged across all possible coalition permutations:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \left[v(S \cup \{i\}) - v(S) \right] \tag{12}$$

2. The Additive Explanation Surrogate Model

SHAP models local explanations via an additive linear surrogate $g(z')$ defined over simplified binary presence indicators $z' \in \{0,1\}^M$:

$$g(z') = \phi_0 + \sum_{i=1}^{M} \phi_i z'_i \quad \text{where} \quad \phi_0 = \mathbb{E}[f(X)] \tag{13}$$

When all features are present ($z' = \mathbf{1}$), the surrogate matches the exact model prediction: $f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i$.

3. The Four Fundamental Axioms

Shapley values are mathematically proven to be the **only** attribution scheme that simultaneously satisfies four essential fairness properties:

Axiom 1. Efficiency: The sum of all attributions equals the total prediction shift from baseline:

$$\sum_{i=1}^{M} \phi_i = f(x) - \mathbb{E}[f(X)] \tag{14}$$

Axiom 2. Symmetry (Equal Treatment): If two features i and j contribute identically to every coalition ($v(S \cup \{i\}) = v(S \cup \{j\}) \quad \forall S \subseteq N \setminus \{i, j\}$), then their attributions are equal: $\phi_i = \phi_j$.

Axiom 3. Dummy / Null Player: If feature i provides zero marginal gain to all coalitions ($v(S \cup \{i\}) = v(S) \quad \forall S$), its attribution is strictly zero: $\phi_i = 0$.

Axiom 4. Additivity / Linearity: If a model is an ensemble sum of two submodels $f(x) = f_1(x) + f_2(x)$, then $\phi_i(f) = \phi_i(f_1) + \phi_i(f_2)$.

4. Explanation Representation Provided

SHAP delivers explanations in two complementary representations:

- **Local Representation:** Signed scalar feature importances visualized via *Waterfall Plots* and *Force Plots*, directly showing which features drove the prediction toward a positive or negative diagnosis relative to the base value ϕ_0.
- **Global Aggregation:** Aggregating local Shapley values across the entire dataset via *Beeswarm Summary Plots* and global importance rankings:

$$I_j = \frac{1}{N} \sum_{k=1}^{N} \left| \phi_j^{(k)} \right| \tag{15}$$

5. Core Advantages for Model Interpretability

Unlike heuristic methods like LIME, SHAP offers **axiomatic consistency** and **interaction sensitivity**: it is mathematically guaranteed that if a model is modified such that a feature's marginal contribution increases, its reported importance cannot decrease (a property frequently violated by other methods). Furthermore, by averaging across all coalitions, it accounts for non-linear multi-feature interactions rather than evaluating features in complete isolation.

★ Examiner Feedback & Common Pitfalls (Prof. Eliana Pastor)

In official grading, students were explicitly penalized (-0.20 points) for failing to identify that SHAP in its foundational formulation (KernelSHAP) is a **model-agnostic** technique. Always state explicitly: "SHAP is a post-hoc, local, model-agnostic feature attribution method."

➥ Course Theory Mapping (Notes & Slides)

NOTES/XAI_Theory.pdf – Chapter 5: "Post-Modeling Explainability", Sections: "Shapley Values for XAI", "SHAP", and "KernelSHAP" (pp. 48–62).

¶ Alternative Perspectives & Methodological Variants

Because exact Shapley calculation requires exponential 2^M evaluations, practical implementations employ specialized approximations: **TreeSHAP** achieves exact computation in $\mathcal{O}(TLD^2)$ for tree ensembles, while **DeepSHAP** adapts DeepLIFT for neural network backpropagation.

Question 6: Clinical Diagnostic System for Medical Images (X-Rays / MRIs) — Open Inquiry — 6.00 Points

A healthcare provider uses an AI system to diagnose diseases from medical images (e.g., X-rays or MRIs). As part of ensuring the system is trustworthy and useful in clinical settings, it is important that the AI provides understandable explanations to doctors.
Describe at least two explainability methods that could be used to help doctors understand the decisions of the system. Discuss the types of explanation representation (e.g., feature importance, local rules, visualization example-based) that would be most appropriate for this scenario and why. Choose one explainability method you would recommend for this use case. Clearly justify your choice based on the needs of healthcare professionals.

✓ Executive Solution Summary

Medical image diagnosis on X-rays/MRIs requires bridging the semantic gap between high-dimensional raw pixel tensors and clinical pathology. Because individual pixels lack semantic meaning, explainability methods **must employ interpretable representations** (superpixels, anatomical bounding boxes, or clinical visual concepts).
We analyze three prominent methods: (1) **Local Feature Attribution via Superpixels** (LIME/SHAP for images), (2) **Gradient-Weighted Class Activation Mapping** (Grad-CAM), and (3) **Concept Bottleneck Models** (CBMs). We discuss four explanation representations: visual saliency heatmaps, concept attributions, example-based prototypes, and counterfactuals. We explicitly **recommend Concept Bottleneck Models (CBMs)**, because they express decision logic entirely in standard medical vocabulary (clinical concepts), guarantee by-design architectural faithfulness, eliminate hallucinated correlations, and empower radiologists to

perform **test-time concept interventions** (overriding misdetected findings to update the final diagnosis dynamically).

✍ **Comprehensive Technical Walkthrough**

1. Clinical Domain Desiderata & The Problem of Image Representation

Medical imaging diagnosis (e.g., detecting pneumonia, cardiomegaly, or tumors) carries zero tolerance for unfaithful explanations due to clinical liability and patient safety.

The Imperative of Interpretable Representations for Image Data

A fundamental technical challenge in computer vision XAI is that raw inputs $x \in \mathbb{R}^{H \times W \times C}$ are high-dimensional tensors of raw pixels. **A single pixel (i, j, c) has zero diagnostic or semantic meaning to a physician.** Therefore, any explainability pipeline operating on medical images must map pixel space into an **interpretable representation** x':

$$h_x : \{0,1\}^M \to \mathbb{R}^{H \times W \times C} \tag{16}$$

Standard interpretable representations include:

- **Superpixels / Image Patches:** Grouping contiguous pixels sharing color and texture via segmentation algorithms (SLIC, Quickshift, Felzenszwalb) into M anatomically coherent regions (e.g., left lung apex, costophrenic angle).
- **Anatomical Bounding Boxes & ROI Masks:** Clinically delineated anatomical zones (e.g., cardiac silhouette, mediastinum).
- **High-Level Clinical Concepts:** Semantic clinical findings (e.g., *"pleural effusion"*, *"lung opacity"*, *"bone fracture"*, *"cardiomegaly"*).

2. In-Depth Analysis of Explainability Methods

Method 1: Local Feature Attribution via Superpixels (LIME for Images)

- **Mechanism:** The radiograph is segmented into M superpixels. LIME generates perturbed binary masks $z' \in \{0,1\}^M$ where subsets of superpixels are grayed out or blurred. The perturbed images $h_x(z')$ are passed through the black-box model to obtain predictions $f(h_x(z'))$. A sparse linear surrogate model $g(z') = w_0 + \sum_{i=1}^{M} w_i z'_i$ is fitted by minimizing:

$$\mathcal{L}(f, g, \pi_x) = \sum_{z'} \pi_x(z') \left(f(h_x(z')) - g(z')\right)^2 + \Omega(g) \tag{17}$$

  where $\pi_x(z') = \exp(-D(x, z')^2/\sigma^2)$ measures proximity to the query image.
- **Clinical Utility:** Highlights which anatomical regions positively or negatively contributed to the diagnostic score.
- **Limitation:** Superpixel boundaries are arbitrary and often slice across pathological lesion contours; vulnerable to sample instability.

Method 2: Gradient-Weighted Class Activation Mapping (Grad-CAM)

- **Mechanism:** Grad-CAM utilizes the penultimate convolutional layer, which retains rich spatial localization alongside high-level semantic abstractions. It computes the gradient of the clinical class score Y^c with respect to feature activation maps A^k:

$$\alpha_k^c = \frac{1}{Z} \sum_{i=1}^{U} \sum_{j=1}^{V} \frac{\partial Y^c}{\partial A_{i,j}^k} \tag{18}$$

  The coarse localization heatmap is generated via a rectified weighted sum:

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right) \tag{19}$$

- **Clinical Utility:** Produces an intuitive visual heatmap overlay on the patient's X-ray, confirming whether the CNN focused on true pathology (e.g., apical nodule) or an external artifact (e.g., hospital marker).
- **Limitation:** Low spatial resolution; cannot distinguish whether activation is due to shape, density, or surrounding tissue.

Method 3: Concept Bottleneck Models (CBMs — By-Design)

- **Mechanism:** The architecture decomposes the diagnostic pipeline into two explicit stages:

$$x \xrightarrow{g} \hat{\mathbf{c}} \xrightarrow{f} \hat{y} \tag{20}$$

  where $g(x)$ predicts K clinically defined radiological concepts (e.g., $\hat{c}_1$ = consolidation, $\hat{c}_2$ = effusion, $\hat{c}_3$ = cardiomegaly), and $f(\hat{\mathbf{c}})$ is an interpretable classifier (such as a sparse linear model or shallow decision tree) mapping concepts to disease probability:

$$P(y = \text{Pneumonia}) = \sigma\left(w_1\hat{c}_1 + w_2\hat{c}_2 + \cdots + w_K\hat{c}_K + b\right) \tag{21}$$

- **Clinical Utility:** Explains diagnoses using exactly the clinical vocabulary taught in medical school.

3. Evaluation of Explanation Representations for Healthcare

Representation Type	Visual / Structural Form	Suitability for Doctors	Primary Failure Mode
Visual Saliency	Heatmap overlay on anatomical image	High immediacy; rapidly guides radiologist's gaze to region	Suffers from semantic gap; cannot explain *why* lesion is abnormal
Concept Attributions	Symbolic list of weighted clinical findings	Maximum clinical alignment; matches diagnostic differential lists	Requires concept-annotated medical datasets
Example-Based	Historical case retrieval / prototypes	High comfort; matches clinical case-based reasoning	Risk of superficial visual similarity masking underlying differences
Counterfactuals	Modified synthetic image showing lesion removal	Highly actionable; demonstrates boundary between benign and malignant	Off-manifold generative artifacts; risk of clinical hallucination

4. Final Recommendation and Rigorous Justification

Recommendation: We formally recommend implementing a **Concept Bottleneck Model (CBM)** (optionally complemented with Grad-CAM overlays on predicted concept regions).

Justification Based on Healthcare Professional Needs:

J1. Alignment with Clinical Reasoning and Medical Lexicon: Physicians do not make diagnoses based on raw pixel clusters or arbitrary superpixels. They reason symbolically via established pathological signs (e.g., "interstitial opacities", "blunted costophrenic angles"). CBMs provide explanations directly in this formal vocabulary.

J2. Human-in-the-Loop Test-Time Concept Intervention: This is the single most critical capability in clinical medicine. If an imaging artifact (such as a thoracic drainage tube or patient jewelry) causes the concept encoder $g(x)$ to misdetect "consolidation", the radiologist can directly intervene on the user interface:

$$\hat{c}_{\text{consolidation}} \leftarrow 0 \implies \hat{y}_{\text{updated}} = \sigma\left(\sum_{j \neq \text{consolidation}} w_j \hat{c}_j + w_{\text{consolidation}} \cdot 0 + b\right) \tag{22}$$

The diagnostic score immediately recalculates without requiring model retraining, empowering physicians to correct AI errors safely.

J3. Architectural Faithfulness (Zero Post-Hoc Hallucination): Unlike post-hoc surrogates (LIME/SHAP) that only approximate local neighborhoods, the linear concept layer in CBM is *inherently faithful by construction*. There is zero risk of an unfaithful explanation leading to medical malpractice.

J4. Auditability and Regulatory Compliance: Satisfies FDA Software-as-a-Medical-Device (SaMD) and EU AI Act requirements for high-risk clinical decision support systems, providing fully traceable, auditable diagnostic logic.

★ Examiner Feedback & Common Pitfalls (Prof. Eliana Pastor)

In official grading, students suffered heavy mark deductions (up to −1.50 points) for two specific oversights:

1. **Omitting the need for interpretable representations for image data:** Candidates must explicitly explain that raw pixels are meaningless to doctors and explain how images are converted to superpixels, bounding boxes, or concepts (−0.75 points).
2. **Missing or weakly justified final recommendation:** Candidates who discussed multiple methods but failed to state an explicit final choice with healthcare-specific justification lost an additional −0.75 points.

➥ Course Theory Mapping (Notes & Slides)

NOTES/XAI_Theory.pdf – Chapter 2: "Taxonomy of Explainable AI" (pp. 12–21), Chapter 5: "Local Surrogate Models & Grad-CAM" (pp. 42–47, 72–76), and Chapter 7: "Concept-based Explainable AI: CBMs" (pp. 98–106).

¶ Alternative Perspectives & Methodological Variants

ProtoPNet (This Looks Like That) provides an alternative by-design approach: it breaks radiographs into prototypical lesion patches, explaining decisions through case-based comparison with verified historical clinical prototypes stored during training.